



The University of Azad Jammu and Kashmir

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Course Title: Database System

Course Code: CS-22034

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LAB 06

1. Introduction

This lab was performed to understand SQL filtering techniques using MySQL in XAMPP. Different SQL operators and clauses such as WHERE, AND, OR, BETWEEN, IN, LIKE, IS NULL, ORDER BY, and LIMIT were used to retrieve specific employee records from the Employee table.

2. Objective

The objective of this lab was to practice SQL filtering and sorting operations on a database table using MySQL queries.

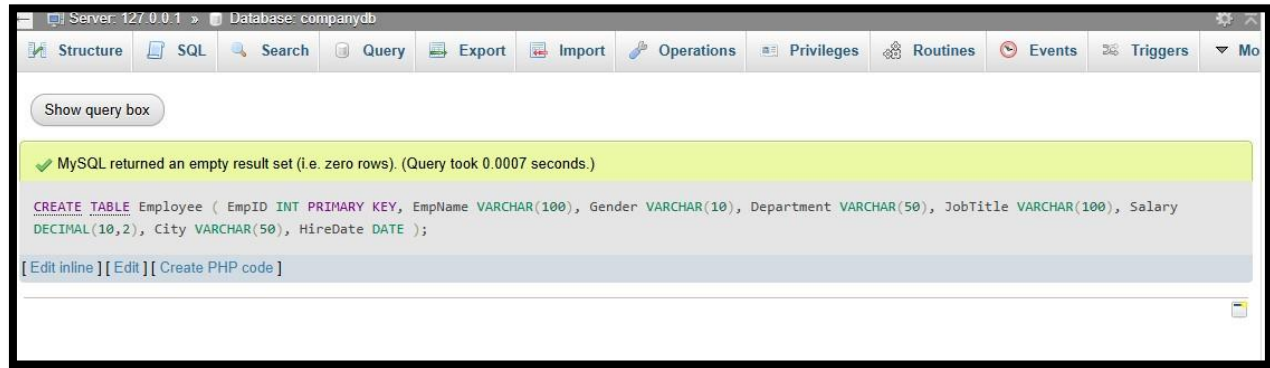
3. Database Creation

First, the XAMPP server was started and phpMyAdmin was opened through localhost. A database named “**companydb**” was created successfully.



4. Employee Table Creation

An Employee table was created in the companydb database using **SQL CREATE TABLE** statement.



Data Insertion

Sample employee records were inserted into the Employee table using INSERT INTO statement.



Employee records inserted successfully.

5. Part A Queries

Task A1

- This query displays employee ID, employee name, and salary of employees earning more than 90,000.

```
SELECT EmpID, EmpName, Salary
FROM Employee
WHERE Salary > 90000;
```

Employees earning more than 90,000.

Showing rows 0 - 3 (4 total, Query took 0.0005 seconds.)

-- Task A1 `SELECT EmpID, EmpName, Salary FROM Employee WHERE Salary > 90000;`

[Edit inline] [Edit] [Create PHP code]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

				EmpID	EmpName	Salary
☐	Edit	Copy	Delete	1	Ali Khan	95000.00
☐	Edit	Copy	Delete	5	Usman Tariq	120000.00
☐	Edit	Copy	Delete	8	Minal Fatima	99000.00
☐	Edit	Copy	Delete	10	Maryam Imran	105000.00

☐ Check all | With selected: Edit Copy Delete Export

Output of Task A1 query.

Task A2

- This query shows employees whose salary is less than or equal to 75,000.

```
SELECT EmpName, Salary
FROM Employee
WHERE Salary <= 75000;
```

Show query box

✓ Showing rows 0 - 1 (2 total, Query took 0.0015 seconds.)

```
-- Task A2 SELECT EmpName, Salary FROM Employee WHERE Salary <= 75000;
```

Edit inline | [Edit] | [Create PHP code]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

				EmpName	Salary
☐	Edit	Copy	Delete	Sara Ahmed	72000.00
☐	Edit	Copy	Delete	Bilal Hussain	68000.00

↑ ☐ Check all With selected: Edit Copy Delete Export

Output of Task A2 query showing employees with salary less than or equal to 75,000.

Task A3

- This query finds employees working in Lahore with salary greater than 90,000.

```
SELECT *  
  
FROM Employee  
WHERE City = 'Lahore'  
AND Salary > 90000;
```

Show query box

✓ Showing rows 0 - 2 (3 total, Query took 0.0009 seconds.)

-- Task A3 `SELECT * FROM Employee WHERE City = 'Lahore' AND Salary > 90000;`

Edit inline][Edit][Create PHP code]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

		EmpID	EmpName	Gender	Department	JobTitle	Salary	City	HireDate
☐	Edit Copy Delete	1	Ali Khan	Male	Engineering	Software Engineer	95000.00	Lahore	2021-03-15
☐	Edit Copy Delete	8	Minal Fatima	Female	Sales	Sales Manager	99000.00	Lahore	2020-12-01
☐	Edit Copy Delete	10	Maryam Imran	Female	Engineering	DevOps Engineer	105000.00	Lahore	2019-05-14

☐ Check all | With selected: Edit Copy Delete Export

Output of Task A3 query showing employees from Lahore earning above 90,000.

Task A4

- This query lists employees belonging to Karachi or Islamabad.

```
SELECT EmpName, City FROM Employee
WHERE City = 'Karachi'
OR City = 'Islamabad';
```

✓ Showing rows 0 - 4 (5 total, Query took 0.0005 seconds.)

```
-- Task A4 SELECT EmpName, City FROM Employee WHERE City = 'Karachi' OR City = 'Islamabad';
```

[Edit inline] [Edit] [Create PHP code]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	EmpName	City
☐ Edit Copy Delete	Sara Ahmed	Karachi
☐ Edit Copy Delete	Bilal Hussain	Islamabad
☐ Edit Copy Delete	Usman Tariq	Karachi
☐ Edit Copy Delete	Daniyal Raza	Islamabad
☐ Edit Copy Delete	Hassan Ali	Karachi

↑ ☐ Check all With selected: Edit Copy Delete Export

Output of Task A4 query showing employees from Karachi or Islamabad.

Task A5

- This query displays female employees who are not part of the Engineering department.

```
SELECT *  
FROM Employee  
WHERE Gender = 'Female'  
AND Department <>  
'Engineering';
```

✓ Showing rows 0 - 2 (3 total, Query took 0.0006 seconds.)

-- Task A5 `SELECT * FROM Employee WHERE Gender = 'Female' AND Department <> 'Engineering';`

[Edit inline] [Edit] [Create PHP code]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	EmpID	EmpName	Gender	Department	JobTitle	Salary	City	HireDate
☐ Edit Copy Delete	2	Sara Ahmed	Female	Marketing	Marketing Officer	72000.00	Karachi	2020-07-10
☐ Edit Copy Delete	4	Maham Noor	Female	HR	HR Manager	88000.00	Lahore	2022-01-20
☐ Edit Copy Delete	8	Minal Fatima	Female	Sales	Sales Manager	99000.00	Lahore	2020-12-01

↑ ☐ Check all With selected: Edit Copy Delete Export

Output of Task A5 query showing female employees outside Engineering department.

Task A6

- This query finds male employees earning between 70,000 and 90,000.

```
SELECT *
FROM Employee
WHERE Gender = 'Male'
AND Salary >= 70000
AND Salary <= 90000;
```

✓ Showing rows 0 - 1 (2 total, Query took 0.0005 seconds.)

```
-- Task A6 SELECT * FROM Employee WHERE Gender = 'Male' AND Salary >= 70000 AND Salary <= 90000;
```

[Edit inline] [Edit] [Create PHP code]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	EmpID	EmpName	Gender	Department	JobTitle	Salary	City	HireDate
☐ Edit Copy Delete	7	Daniyal Raza	Male	Finance	Accountant	76000.00	Islamabad	2021-06-25
☐ Edit Copy Delete	9	Hassan Ali	Male	Marketing	SEO Specialist	81000.00	Karachi	2022-08-30

☐ Check all | With selected: Edit Copy Delete Export

Output of Task A6 query showing male employees earning between 70,000 and 90,000.

Task A7

- This query lists employees who are Software Engineers or have salary greater than 100,000.

```
SELECT *
FROM Employee
WHERE JobTitle = 'Software
Engineer'
OR Salary > 100000;
```

✓ Showing rows 0 - 2 (3 total, Query took 0.0006 seconds.)

-- Task A7 `SELECT * FROM Employee WHERE JobTitle = 'Software Engineer' OR Salary > 100000;`

[Edit inline] [Edit] [Create PHP code]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

		EmpID	EmpName	Gender	Department	JobTitle	Salary	City	HireDate
☐	Edit Copy Delete	1	Ali Khan	Male	Engineering	Software Engineer	95000.00	Lahore	2021-03-15
☐	Edit Copy Delete	5	Usman Tariq	Male	Engineering	Senior Software Engineer	120000.00	Karachi	2018-09-12
☐	Edit Copy Delete	10	Maryam Imran	Female	Engineering	DevOps Engineer	105000.00	Lahore	2019-05-14

↑ ☐ Check all | With selected: Edit Copy Delete Export

Output of Task A7 query showing Software Engineers or employees earning above 100,000.

Task A8

- This query excludes employees from Marketing and Sales departments.

```
SELECT *
FROM Employee
WHERE Department <>
'Marketing'
AND Department <> 'Sales';
```

✓ Showing rows 0 - 5 (6 total, Query took 0.0005 seconds.)

-- Task A8 `SELECT * FROM Employee WHERE Department <> 'Marketing' AND Department <> 'Sales';`

[Edit inline](#) [\[Edit \]](#) [\[Create PHP code \]](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

				EmpID	EmpName	Gender	Department	JobTitle	Salary	City	HireDate
☐				1	Ali Khan	Male	Engineering	Software Engineer	95000.00	Lahore	2021-03-15
☐				4	Maham Noor	Female	HR	HR Manager	88000.00	Lahore	2022-01-20
☐				5	Usman Tariq	Male	Engineering	Senior Software Engineer	120000.00	Karachi	2018-09-12
☐				6	Areeba Khan	Female	Engineering	QA Engineer	85000.00	NULL	2023-04-18
☐				7	Daniyal Raza	Male	Finance	Accountant	76000.00	Islamabad	2021-06-25
☐				10	Maryam Imran	Female	Engineering	DevOps Engineer	105000.00	Lahore	2019-05-14

☐ Check all | With selected: Edit Copy Delete Export

Output of Task A8 query excluding Marketing and Sales employees.

6. Conclusion

In this lab, different SQL filtering and sorting operations were performed successfully using MySQL in XAMPP. Various SQL clauses such as WHERE, AND, OR, BETWEEN, IN, LIKE, IS NULL, ORDER BY, and LIMIT were used to retrieve and manage employee records from the Employee table. This lab improved understanding of SQL query writing, data filtering techniques, pattern matching, and result sorting in.

*****THE END*****